

Customer Interface Box Circuit Breakers Tripped

Symptoms

- Circuit breakers are frequently tripping on the engine customer interface box (C.I.B.).

How To Use This Tree

This symptom tree can be used to troubleshoot breaker tripping on the C.I.B. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Verify battery disconnect switch is turned ON.

Possible causes:

- Keyswitch circuit is shorted
- Backfeed from alternator
- Both isolated and non-isolated systems are on option. Be aware of both.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check fault codes.	
	STEP 1A. Check for active fault codes.	Any fault codes active?
	STEP 1B. Check the engine interface wiring harness.	30 ampere breaker trip?
	STEP 1C. Check the engine interface wiring harness.	10 ampere breaker trip?
STEP 2.	Check the display adapter wiring harness.	
	STEP 2A. Inspect the wiring harness connector pins.	Dirty or damaged pins?
	STEP 2B. Check for a pin-to-pin short in the wiring harness.	Less than 10 ohms?

STEPS		SPECIFICATIONS
STEP 3.	Check the helm harness.	
	STEP 3A. Inspect the wiring harness connector pins.	Dirty or damaged pins?
	STEP 3B. Check for a pin-to-pin short in the wiring harness.	Greater than 100k ohms?
STEP 4.	Check the main extension harness and engine control module (ECM).	
	STEP 4A. Inspect the wiring harness connector pins.	Dirty or damaged pins?
	STEP 4B. Check for a pin-to-pin short in the wiring harness.	Greater than 100k ohms?
STEP 5.	Check the engine interface wiring harness.	
	STEP 5A. Inspect the wiring harness connector pins.	Dirty or damaged pins?
	STEP 5B. Check for a pin-to-pin short in the wiring harness.	Greater than 100k ohms?
STEP 6.	Check the Engine Interface wiring harness.	
	STEP 6A. Inspect the wiring harness connector pins.	Dirty or damaged pins?
	STEP 6B. Check for a pin-to-pin short in the wiring harness.	Greater than 100k ohms?

STEP 1. Check fault codes.

STEP 1A. Check for active fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Check for active fault codes. Use INSITE™ electronic service tool to read the fault codes.	Any fault codes active? YES	Appropriate fault code troubleshooting tree
	Any fault codes active? NO	1B

STEP 1B. Check the engine interface wiring harness.

Conditions :

- Turn system enable switch OFF.

Action	Specification/Repair	Next Step
Check the engine interface wiring harness. Verify the status of the circuit breakers on the C.I.B.	30 ampere breaker trip? YES	5A
	30 ampere breaker trip? NO	1C

STEP 1C. Check the engine interface wiring harness.

Conditions :

- Turn system enable switch OFF.

Action	Specification/Repair	Next Step
Check the engine interface wiring harness. Verify the status of the circuit breakers on the C.I.B.	10 ampere breaker trip? YES	2A
	10 ampere breaker trip? NO	Repair complete

STEP 2. Check the display adapter wiring harness.

STEP 2A. Inspect the wiring harness connector pins.

Conditions :

- Turn system enable switch OFF.
- Disconnect the helm harness from the display adapter wiring harness.
- Disconnect the ED-4 display from the display adapter wiring harness.

Action	Specification/Repair	Next Step
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Inspect the wiring harness connector pins. • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the display adapter harness or the display adapter harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. • Refer to Procedure 015-106 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-106.html)	Repair complete.
	Dirty or damaged pins? NO	2B

STEP 2B. Check for a pin-to-pin short in the wiring harness.

Conditions :

- Turn system enable switch OFF.
- Disconnect the helm harness from the display adapter wiring harness.
- Disconnect the ED-4 display from the display adapter wiring harness.

Action	Specification/Repair	Next Step
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Check for a pin-to-pin short in the wiring harness. Measure resistance between the keyswitch in the display adapter wiring harness connector and all other pins in the display adapter wiring harness connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	3A
	Greater than 100k ohms? NO Repair : Replace the faulty wire. Refer to Procedure 015-106 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-106.html)	Repair complete

STEP 3. Check the helm harness.

STEP 3A. Inspect the wiring harness connector pins.

Conditions :

- Turn system enable switch OFF.
- Disconnect all harnesses connecting to the helm wiring harness.

Action	Specification/Repair	Next Step

Inspect the wiring harness connector pins. • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the helm connector or the helm harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. • Refer to Procedure 015-078 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-078.html)	Repair complete
	Dirty or damaged pins? NO	3B

STEP 3B. Check for a pin-to-pin short in the wiring harness.

Conditions :

- Turn system enable switch OFF.
- Disconnect all harnesses connecting to the helm wiring harness.

Action	Specification/Repair	Next Step
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Check for a pin-to-pin short in the wiring harness. Measure resistance between the keyswitch in the helm wiring harness connector and all other pins in the helm wiring harness connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	4A
	Greater than 100k ohms? NO Repair : Replace the faulty wire. Refer to Procedure 015-078 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-078.html)	Repair complete

STEP 4. Check the main extension harness and engine control module (ECM).

STEP 4A. Inspect the wiring harness connector pins.

Conditions :

- Turn system enable switch OFF.
- Disconnect all harnesses connecting to the main extension wiring harness.

Action	Specification/Repair	Next Step

Inspect the wiring harness connector pins. • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. • Refer to Procedure 015-077 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-077.html)	Repair complete
	Dirty or damaged pins? NO	4B

STEP 4B. Check for a pin-to-pin short in the wiring harness.

Conditions : • Turn system enable switch OFF. • Disconnect helm harnesses from the main extension wiring harness. • Disconnect the main extension harness from the C.I.B.		
Action	Specification/Repair	Next Step

Check for a pin-to-pin short in the wiring harness. Measure resistance between the keyswitch in the main extension wiring harness connector and all other pins in the main extension wiring harness connector.	Greater than 100k ohms? YES Repair : Replace the C.I.B. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-023.html)	Repair complete
Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? NO Repair : Replace the faulty wire. Refer to Procedure 015-077 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-077.html)	Repair complete

STEP 5. Check the engine interface wiring harness.

STEP 5A. Inspect the wiring harness connector pins.

Conditions :

- Turn system enable switch OFF.
- Disconnect engine interface wiring harness from the C.I.B.
- Disconnect engine interface wiring harness from the original equipment manufacturer (OEM) interface panel.

Action	Specification/Repair	Next Step

Inspect the wiring harness connector pins. • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the main extension connector or the main extension harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. • Refer to Procedure 015-093 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-093.html)	Repair complete
	Dirty or damaged pins? NO	5B

STEP 5B. Check for a pin-to-pin short in the wiring harness.

Conditions : • Turn system enable switch OFF. • Disconnect engine interface wiring harness from the C.I.B. • Disconnect engine interface wiring harness from the OEM interface panel.		
Action	Specification/Repair	Next Step

Check for a pin-to-pin short in the wiring harness. Measure resistance between the keyswitch pin 33 in the engine interface wiring harness connector and all other pins in the engine interface wiring harness connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	6A
	Greater than 100k ohms? NO Repair : Replace the faulty wire. Refer to Procedure 015-093 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-093.html)	Repair complete

STEP 6. Check the engine interface wiring harness.

STEP 6A. Inspect the wiring harness connector pins.

Conditions :

- Turn system enable switch OFF.
- Disconnect engine interface wiring harness from the OEM interface panel.

Action	Specification/Repair	Next Step

Inspect the wiring harness connector pins. • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the main extension connector or the main extension harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. • Refer to Procedure 015-093 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-093.html)	Repair complete
	Dirty or damaged pins? NO	6B

STEP 6B. Check for a pin-to-pin short in the wiring harness.

Conditions :

- Turn system enable switch OFF.
- Disconnect engine interface wiring harness from the OEM interface panel.

Action	Specification/Repair	Next Step
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Check for a pin-to-pin short in the wiring harness. • Measure resistance between the keyswitch pin 33 on the engine wiring harness OEM interface connector and all other pins in the OEM interface connector.	Greater than 100k ohms? YES Repair : Replace the C.I.B.. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-023.html)	Repair complete
Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? NO Repair : Replace the faulty wire. • Refer to Procedure 015-093 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-093.html)	Repair complete